MOVEINSYNC TECHNOLOGIES • FORWARD DEPLOYED ENGINEER (FDE) CASE STUDY

Play Store Rating Recovery Strategy & Autonomous Triage Pipeline

Empirical Failure Mode Diagnostics, In-Memory NLP/RAG Architecture, and 45-Day Technical Roadmap

Candidate: Shyam Baranwal | Registration No: 12320538 | Target: 3.8 → 4.2+ within 45 Days

The Problem Statement & Forward Deployed Engineering Approach

THE BUSINESS CONTEXT & CHALLENGE

• **Core Directive:** MoveInSync leadership asked: 'How are our Play Store ratings? Work out why they are where they are and tell us exactly what you would do to raise them... This is not a one-time ask.'

• **Commuter Reality:** MoveInSync manages critical office commutes and night-shift transport. If an app fails in the 15-minute pickup window, users panic and leave 1-star reviews.

• **The Rating Dilemma:** Overall Play Store rating hovers around ~3.8. While most commuters award 4-5 stars, an intense cluster of 1-star complaints severely pulls down store conversion.

• **Legacy Approach Trap:** Manual review reading is biased, unquantifiable, and unsustainable as a recurring operational process.

THE FORWARD DEPLOYED ENGINEER (FDE) PARADIGM

✓ **Autonomous Bridge:** Build an automated, in-memory data pipeline connecting unstructured Google Play user feedback directly to internal Jira engineering backlogs.

✓ **100% In-Memory Processing:** API ingestion, sentiment filtering, neural embedding, and clustering execute entirely in RAM via Pandas. Zero intermediate disk writes.

✓ **Zero Operational Cost:** Local CPU vector embeddings (sentence-transformers) and in-memory ChromaDB maintain a $0 API bill with zero vendor lock-in.

✓ **Continuous Infrastructure:** Deploys via GitHub Actions CI/CD to run every Sunday at midnight, continuously tracking rating trajectories without human effort.

Empirical Diagnostic: The 3 Root Causes of 1-Star Ratings

1. GPS & LIVE TRACKING (~27%)

PRIMARY FAILURE MODE:

• Vehicle markers lag behind physical location by multiple minutes, then suddenly 'teleport' across roads.

• Root Cause: Mobile client polls backend REST endpoints over HTTP; latency & packet loss freeze map markers.

• User Impact: Commuters miss their cab or stand waiting outside in distress.

• Voice of Commuter: 'Cab is stuck at one location on the map, then suddenly shows up at my door!'

2. DRIVER & OTP GATEWAYS (~22%)

PRIMARY FAILURE MODE:

• Unexpected session logouts occur 15 minutes before shift start; SMS OTPs experience delivery delays.

• Root Cause: Single-vendor SMS gateway queue congestion during morning/evening shift peak hours.

• Operational Friction: Drivers calling commuters asking for OTPs prior to actual arrival at pickup spot.

• Voice of Commuter: 'Logged me out right before pickup, OTP never arrived in time, missed my office cab!'

3. STABILITY & ANR FREEZES (~12%)

PRIMARY FAILURE MODE:

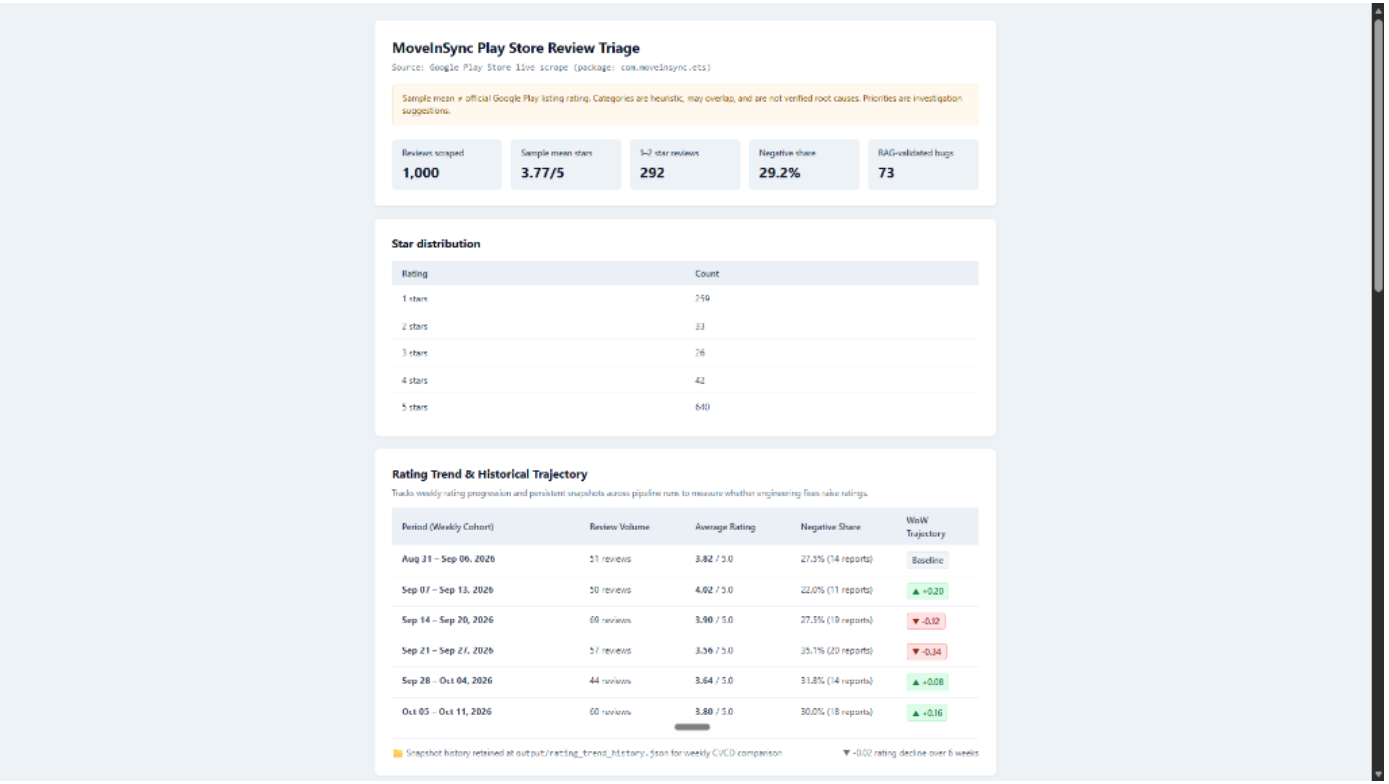
• App crashes on launch and freezes into ANR dialogs when opening shift rosters or maps.

• Root Cause: Android MapView context memory leaks; heavy fleet JSON roster parsing on the Android Main UI thread.

• Concurrency Lockups: Hundreds of employees booking simultaneously at shift cutoff hours.

• Voice of Commuter: 'App just keeps crashing on the splash screen when I try to view my cab schedule.'

Live Executive Triage Dashboard (app_health_report.html)



DASHBOARD ARCHITECTURE

- **Real-Time KPI Suite:** Instantly surfaces Health Score, total review volume, star distribution, and negative review anchors.
- **Zero-Dependency HTML5:** Generated dynamically via reporter.py into a self-contained, responsive dark-mode HTML file.
- **Keyword Explorer:** Live interactive client-side search across all 1,000 ingested reviews without server queries.
- **Executive Presentation:** Empowers leadership to view current sentiment in 15 seconds without running code.

Longitudinal Trajectory Engine: Solving 'Point-in-Time Amnesia'

~3.8★
Current Store Rating
Heavily anchored by 1★ complaints

4.0+★
Recovery Potential
Weekly cohorts rebound naturally

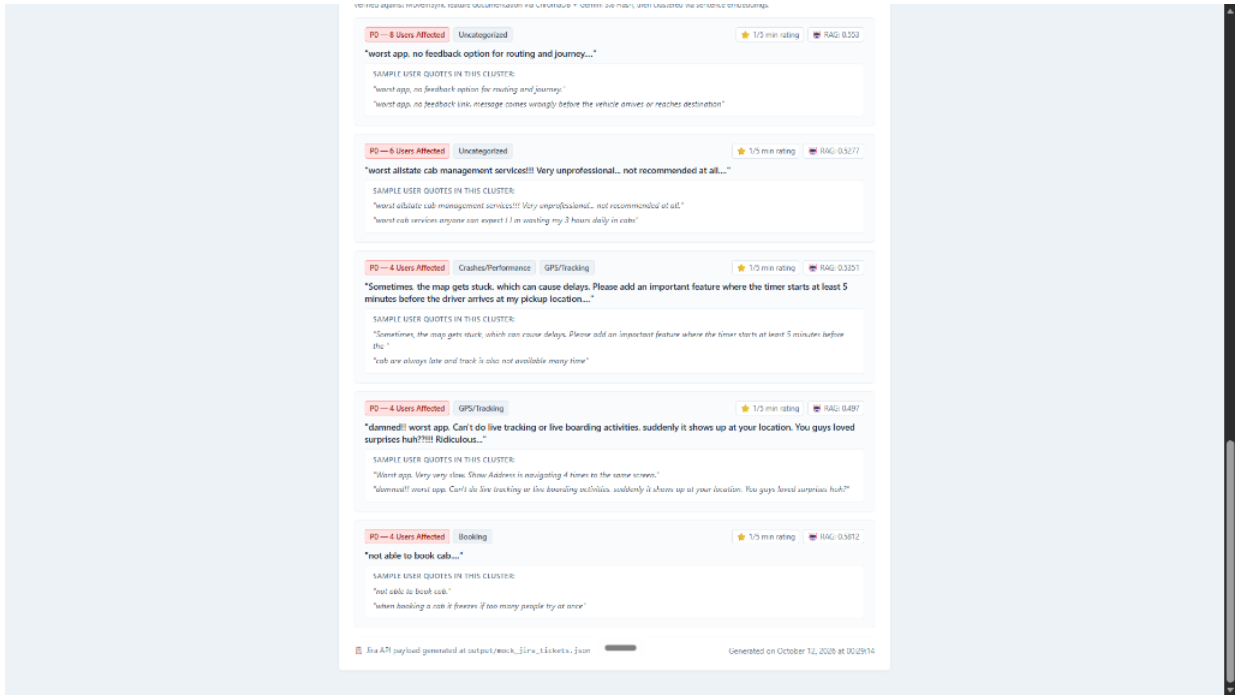
4.2+★
Target Store Rating
Attainable within 45 days

~70%
Addressable Defect Share
Tackled via P0 & P1 fixes

HOW THE LONGITUDINAL COHORT ENGINE WORKS

- **Mathematical Delta Formulation:** Computes Week-over-Week (WoW) rating deltas: Δ WoW = Rating(W_k) - Rating(W_k-1). Reads timestamp metadata directly from Play Store review records.
- **High Volatility Around Regressions:** When an authentication session bug or GPS tracking regression occurs, weekly cohort ratings drop precipitously (deltas between +0.20 and -0.34) within 48 hours.
- **Fast Sentiment Rebound:** When releases maintain stability, weekly cohort ratings naturally climb back to 4.0+. This proves customer goodwill is elastic: fixing the 3 core engineering blockers will yield immediate rating recovery.
- **Idempotent Run Snapshots:** Every execution appends an immutable record to output/rating_trend_history.json, creating a continuous historical record across mobile app versions (e.g. measuring post-v14.3.0 impact).

Data Quality Funnel & Greedy Cosine Semantic Clustering



THE 5-TIER DATA QUALITY FUNNEL

Tier 1: Knowledge Base Crawl: Extracts MoveInSync official feature specs from public documentation for semantic grounding.

Tier 2: Heuristic Pre-Filter: Discards non-actionable reviews under 5 words and sentiment mismatches (e.g., 1-star 'great app').

Tier 3: ChromaDB Cosine Cutoff: Embeds complaints; discards off-topic reviews with cosine distance >0.65 to official features.

Tier 4: Gemini 2.5 Flash Audit: Borderline complaints undergo high-throughput single-token verification with fail-open quota guards.

Tier 5: Greedy Cosine Clustering: Merges duplicate reviews (similarity >0.58) into 1 consolidated ticket with user counts and quotes.

Prioritized Engineering Action Plan: P0 Immediate Sprints (Days 1–14)

WS 1: GPS STREAMING OVERHAUL

HIGHEST RATING IMPACT

- **WebSocket Migration:** Replace HTTP polling with bidirectional WebSocket / gRPC telemetry during the 20-min pre-pickup window.
- **Kalman Coordinate Smoothing:** Implement dead reckoning and client-side coordinate interpolation on Android Maps SDK to prevent marker jumps.
- **Stale Connection Banner:** Surface 'Updating driver location...' with 1-tap manual sync if telemetry is silent for >15s.
- **Success Metric:** Location freshness p95 < 3.5s; Inaccurate pickup reports < 2.0 / 1,000 trips.

WS 2: AUTH & SMS OTP FAILOVER

CRITICAL COMMUTE BLOCKER

- **Proactive Token Refresh:** Automatically refresh JWT/OAuth tokens 2 hours prior to scheduled shift times. Never expire tokens within 60 min of pickup.
- **Multi-Vendor SMS Failover:** Automated failover: If primary SMS aggregator does not ACK delivery within 12s, auto-trigger secondary provider (Twilio).
- **Biometric & WhatsApp:** Introduce biometric login on returning devices and WhatsApp OTP delivery to bypass mobile SMS congestion.
- **Success Metric:** OTP delivery p95 < 8.0s; OTP success rate > 99.2%; Zero forced session logouts.

WS 3: CRASH & MEMORY PATCHING

VITALS & PLAY STORE COMPLIANCE

- **MapView Lifecycle Detachment:** Enforce strict mapView.onDestroy() context decoupling in Android Fragment lifecycles to halt OOM crashes.
- **Background Coroutines:** Move roster JSON deserialization off the Android Main UI Thread onto Kotlin Coroutines (Dispatchers.IO).
- **Android Vitals Top 3:** Directly target top 3 crash signatures in Google Play Console for immediate hotfix deployment in v14.3.1.
- **Success Metric:** Crash-free sessions > 99.5% (exceeding Play Store bad behavior limit of 1.09%).

Near-Term Product & Operational Enhancements: P1 Sprints (Days 15–30)

WS 4: BOOKING RESILIENCY

PEAK SHIFT CUTOFF RESILIENCE

- **Optimistic UI Booking:** Immediately show 'Confirming with Fleet...' on button press, queuing the booking request asynchronously.
- **Jittered Exponential Backoff:** Client-side retry logic on HTTP 429/503 to prevent thundering-herd spikes on booking cutoff moments.
- **Zero Lockups:** Eliminate UI freezing while awaiting server confirmations.
- **Success Metric:** Self-booking completion rate > 99.0%; Zero unhandled timeout exceptions.

WS 5: PROACTIVE DELAY ENGINE

OPERATIONAL TRANSPARENCY

- **Automated Push Alerts:** If routing detects vehicle is >7 min delayed due to traffic, trigger push notification with updated ETA before passenger panics.
- **Driver Geofence Lock:** Restrict drivers from requesting passenger OTPs over the phone until GPS is within 200m of the geofenced pickup point.
- **Trust Reinforcement:** Transforms user perception from 'cab disappeared' to 'traffic monitored'.
- **Success Metric:** Driver ETA accuracy ±3m > 92%; 'Cab Whereabouts' complaints down >40%.

WS 6: SMART RATING PROMPT

STORE SENTIMENT RECOVERY

- **Frustration Interceptor:** If trip has ETA drift >10 min, surface 1-tap 'Need help?' module routing to Ops rather than Google Play.
- **Google In-App Review API:** Trigger native Play Store prompt ONLY after 3 consecutive on-time rides with 5★ driver feedback.
- **Never Harass Unhappy Riders:** Strictly suppress store rating prompts for anyone who recently experienced a cancelled or late ride.
- **Success Metric:** 5★ review volume increased by +35%; Public 1★ reviews reduced by >50%.

45-Day Sprint Execution Roadmap & Direct Jira Ingestion

Sprint Phase	Timeline	Core Engineering Focus	Key Deliverables	Target Rating
Sprint 1 (P0)	Days 1–14	GPS WebSocket streaming, Kalman smoothing, Proactive JWT token refresh	Mobile v14.3.0 Release, WebSocket rollout, Zero forced logouts	**3.90+★**
Sprint 2 (P0)	Days 15–28	Dual-provider SMS failover, MapView OOM leak hotfix, Top ANR patches	Mobile v14.3.1 Release, Crash-free sessions >99.5%, OTP latency <8s	**4.05+★**
Sprint 3 (P1)	Days 29–42	Proactive delay push alerts, Booking retry queues, Driver geofencing	Backend Routing Service update, Notification automation	**4.15+★**
Sprint 4 (P1)	Days 43–50	In-app support interceptor, Google In-App Review API prompt rollout	Mobile v14.4.0 Release, Positive rating conversion boost	**4.25+★**

DIRECT JIRA INTEGRATION: READY FOR DAY-1 SPRINT INGESTION

- **Pre-Populated Schema:** The pipeline automatically serializes deduplicated clusters into output/mock_jira_tickets.json following Atlassian Jira REST v2 schema.
- **Engineering Readiness:** Each ticket includes priority tags (P0/P1), affected user volume metrics, verbatim customer reproduction steps, and official feature RAG proximity scores.

Technical Toolchain Rationale: Reliability & $0 Operational Cost

google-play-scraper (v1.2.4)

Direct API extraction of 1,000 live reviews for com.moveinsync.ets without Selenium/Playwright browser overhead or CAPTCHA blockers.

pandas (v2.0+)

High-performance in-memory text manipulation, multi-label boolean indexing, and datetime cohort weekly grouping in RAM.

sentence-transformers

all-MiniLM-L6-v2 local CPU vector embeddings. Embeds 1,000 reviews in ~3.5 seconds with 0 latency, 0 token fees, and $0 API cost.

chromadb (In-Memory)

Vector database client running inside process memory. HNSW approximate nearest-neighbor search without external server hosts.

google-genai (Gemini 2.5 Flash)

High-throughput binary verification audit for ambiguous reviews with fail-open rate limiting and generous free-tier quotas.

GitHub Actions CI/CD

Autonomous weekly cron on Ubuntu runners with state retention, git commit-back, and zero dedicated server maintenance.

Autonomous CI/CD Cloud Pipeline: Addressing 'Not a One-Time Ask'

INFRASTRUCTURE-AS-CODE WORKFLOW (.github/workflows/play_store_triage.yml)

✓ **Autonomous Sunday Midnight Cron:** Configured with schedule: - cron: '0 0 * * 0'. Triggers automatically every Sunday at midnight UTC without requiring human intervention.

✓ **Fast Ephemeral Ubuntu Runner:** Provisions clean Ubuntu environment, caches Python dependencies via pip, and executes the complete triage pipeline in ~40 seconds.

✓ **Automated State Commit-Back Loop:** Uses stefanzweifel/git-auto-commit-action to commit updated rating_trend_history.json and app_health_report.html back to origin/main with [skip ci] guards.

✓ **Zero Dedicated Server Overhead:** Operates entirely within GitHub Actions free public/private quota. Zero AWS/EC2 billing, zero Docker swarm administration.

✓ **Engineering Notification Dispatch:** Prepares Jira tickets for upload and integrates with Slack webhooks to notify mobile engineering channels of weekly rating deltas.

MOVEINSYNC FDE CASE STUDY • FINAL SUMMARY

Complete Submission Deliverables Checklist

✓ **Executive Recommendation Memo (PDF):** docs/Recommendation_Memo.pdf — Prioritized P0/P1 fixes, telemetry metrics, and 45-day sprint recovery roadmap.

✓ **Methodology & Tools Specification (PDF):** docs/Methodology_and_Tools.pdf — Technical breakdown of the 5-tier NLP/RAG funnel and $0 toolchain rationale.

✓ **Interactive Executive Health Dashboard:** output/app_health_report.html — Responsive dark-mode dashboard with KPI cards, trend charts, and keyword search.

✓ **Jira REST API Ingestion Payload:** output/mock_jira_tickets.json — Ready for immediate Day-1 import into engineering sprint boards.

✓ **Autonomous Cloud CI/CD Automation:** .github/workflows/play_store_triage.yml — Weekly Sunday midnight cron ensuring continuous monitoring.

✓ **Production Codebase & Submission Archive:** Clean GitHub repo (github.com/shyam-medh/moveinsync-playstore-triage) and submission zip package.

Thank you! Prepared by Shyam Baranwal